BUILD SPECIFICATION · DATA LAYER

Legal Soft Lead Pipeline

Twenty-four components across four planes, one relational store, and a lead lifecycle with sixteen states. Discovery runs once per firm; the signal plane runs forever. Everything downstream is a pure function of what those two write.

Scope data layer only **Store** Postgres 16 **Components** D×3 P×9 W×7 R×5 **Lead states** 16

§1 Architecture

Four planes write to one store. The distinction that shapes everything: **discovery and profiling run once per firm and terminate; the signal plane never terminates.** Conflating them is how these systems end up re-crawling the universe nightly.

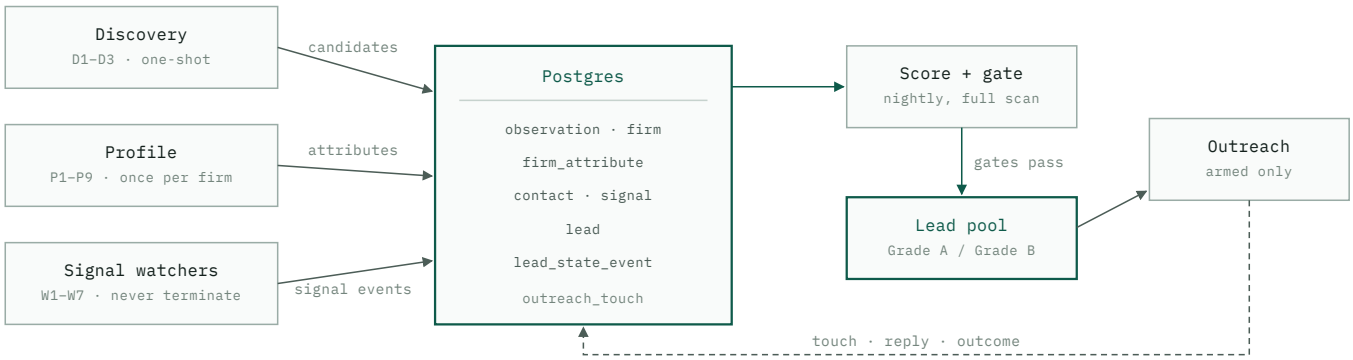


Fig 1 — Every plane writes; nothing reads across planes. Scoring is a nightly full scan of the store, not a stream, so a scoring change is a re-run rather than a backfill. The dashed edge is the only path by which outreach affects targeting.

> THE INVARIANT THAT KEEPS THIS SIMPLE
>
> No component reads another component's output directly. **Everything is mediated by the store.** A scraper that needs a firm's domain reads `firm.domain`; it never calls the discovery scraper. This makes every component independently re-runnable, independently testable against a fixture, and safe to backfill.

§2 ICP as executable predicates

The ICP is not a description, it is a function from firm attributes to a tier. The primary axis is **fee model** — how the firm bills determines whether support staff is a cost centre (sellable) or billable inventory (not sellable at the same margin). Practice area is the secondary axis.

```
-- tier assignment. deterministic, recomputed on any attribute change.
CREATE FUNCTION icp_tier(f firm) RETURNS tier_t AS $$
  SELECT CASE
    -- hard disqualifiers, evaluated first
    WHEN f.entity_kind <> 'private_firm'      THEN 'DQ_ENTITY'
    WHEN f.country <> 'US'                    THEN 'DQ_GEO'
    WHEN f.attorney_count > 50                 THEN 'DQ_SIZE'
    WHEN f.has_inhouse_recruiter                THEN 'DQ_SELFSERVE'
    -- no delegable procedural volume — fails need condition 1
    WHEN f.primary_practice IN ('appellate','patent_prosecution',
                                'complex_commercial')          THEN 'DQ_NOVOLUME'
    WHEN f.first_seen_bar_admission > now() - interval '24 months'
         AND f.attorney_count = 1
         AND NOT f.paid_ads_active              THEN 'DQ_NOBUDGET'

    -- tiers. fee_model gates before practice_area is consulted.
    WHEN f.fee_model IN ('contingency','fee_shifting')
     AND f.primary_practice IN ('personal_injury','mass_tort')      THEN '1A'

    WHEN f.fee_model = 'flat'
     AND f.primary_practice = 'immigration'                          THEN '1B'

    WHEN f.fee_model IN ('contingency','fee_shifting')
     AND f.primary_practice IN ('employment','lemon_law',
                                'workers_comp','ssdi','consumer')       THEN '2'

    WHEN f.fee_model = 'flat'
     AND f.primary_practice IN ('bankruptcy','criminal',
                                'estate','family')                      THEN '3'

    -- hourly firms: only non-billable SKUs are sellable. narrow, not excluded.
    WHEN f.fee_model = 'hourly'                                     THEN '4_NONBILLABLE'

    ELSE 'UNCLASSIFIED'   -- routes to P9 escalation, never to outreach
  END
$$ LANGUAGE sql IMMUTABLE;
```

SIZE POLICY — DISQUALIFY AT 50, PENALISE FROM 26

Firms of 26–50 attorneys are **not** disqualified. They usually have an administrator and sometimes a recruiter, so need condition 4 is weaker and the cycle is longer — but the need is real. They are scored with a −10 fit penalty in §9, which lets a strong signal still surface one, and keeps a whole band from disappearing on an assumption.

The reasoning behind every clause above is in *ICP + Collection Strategy* §1 (the four need conditions), §4 (size) and §6 (disqualifiers). If a rule here and a rule there ever disagree, that document is the source of the *why* and this one is the source of the *how* — reconcile, do not pick.

WEDGE SKU PER TIER

The tier does not just decide whether to contact a firm — it decides which product opens the conversation. This mapping is a lookup table, not logic, so it can be tuned from outcome data without a deploy.

TIER → WEDGE_SKU · SEEDED VALUES, TUNED FROM CLOSE DATA

TIER	SEGMENT	WEDGE SKU	CLIENT-FACING?	SECOND SKU
1A	PI & mass tort	**demand_writer**	NO	document_collector
1B	Immigration	**legal_assistant_es**	PARTLY	document_collector
2	Fee-shifting litigation	**case_manager**	YES	legal_assistant
3	Flat-fee volume	**legal_assistant**	PARTLY	bookkeeper
4	Hourly — non-billable only	**bookkeeper**	NO	marketing_specialist

> **KEEP TIER 4 IN THE SYSTEM**
> Hourly firms are the weakest fit under the fee-model axis. Excluding them from the database entirely makes the axis unfalsifiable — they are never contacted, so no contrary evidence can ever arrive. Classify them, hold them, and route a deliberate holdout slice to outreach so the hypothesis can be measured against real close rates.

§3 Core data structures

Two distinctions carry the whole model. **A `firm` is a durable entity; a `lead` is a time-bounded candidacy for outreach** — one firm produces many leads over its life. And **every attribute claim is stored separately from the resolved value**, because four sources will disagree about a firm's phone number and the pipeline needs to be able to explain which one won.

```
-- ═══ ENTITY ═══════════════════════════════════════════

CREATE TABLE firm (
  id               uuid PRIMARY KEY,
  -- resolved values, materialized from firm_attribute by reconcile()
  canonical_name   text        NOT NULL,
  name_stem        text        NOT NULL,   -- see §4
  domain           text,                   -- eTLD+1, null if no site
  phone_e164       text,
  addr_hash        text,                   -- sha1(street_norm || zip5)
  state            char(2), city text, lat numeric, lng numeric,

  entity_kind      text,       -- private_firm | inhouse | gov | directory_stub
  attorney_count   int,
  staff_ratio      numeric,    -- support headcount / attorney count
  primary_practice text,
  practice_areas   text[],
  fee_model        text,       -- contingency|fee_shifting|flat|hourly|mixed|unknown
  spanish_intake   bool,
  cms_detected     text,
  review_count     int,        -- case-volume proxy
  paid_ads_active  bool,
  has_inhouse_recruiter bool,

  tier             text,       -- output of icp_tier()
  fit_score        int,
  first_seen       timestamptz NOT NULL,
  last_crawled     timestamptz,
  content_hash     text        -- last homepage hash; drives re-extract
);
CREATE INDEX ON firm USING gin (name_stem gin_trgm_ops);
CREATE UNIQUE INDEX ON firm (domain) WHERE domain IS NOT NULL;
CREATE INDEX ON firm (tier, state) WHERE tier NOT LIKE 'DQ%';


-- ═══ PROVENANCE ═══════════════════════════════════════

-- raw, per-source, never mutated. the audit floor.
CREATE TABLE observation (
  id          bigserial PRIMARY KEY,
  source      text NOT NULL,       -- 'places' | 'bar_ca' | 'site' | 'justia'
  source_key  text NOT NULL,       -- place_id, bar number, url
  firm_id     uuid REFERENCES firm,  -- null until resolved
  payload     jsonb NOT NULL,
  fetched_at  timestamptz NOT NULL,
  UNIQUE (source, source_key, fetched_at)
);

-- every claim any source made about any field. resolution reads this.
CREATE TABLE firm_attribute (
  firm_id     uuid NOT NULL REFERENCES firm,
  attr        text NOT NULL,       -- 'phone_e164' | 'fee_model' | ...
  value       jsonb NOT NULL,
  source      text NOT NULL,
  confidence  numeric NOT NULL,    -- 0..1
  evidence    text,                   -- the span the value was read from
  observed_at timestamptz NOT NULL,
  PRIMARY KEY (firm_id, attr, source, observed_at)
);
```

```sql
-- merge audit. lets a bad dedupe be unpicked instead of re-run.
CREATE TABLE firm_alias (
  firm_id     uuid NOT NULL REFERENCES firm,
  alias_name  text, alias_domain text, alias_phone text,
  source      text, merge_rule text, merged_at timestamptz
);


-- ═══ PEOPLE ═══════════════════════════════════════════

CREATE TABLE contact (
  id          uuid PRIMARY KEY,
  firm_id     uuid NOT NULL REFERENCES firm,
  full_name   text NOT NULL,
  title       text,
  role        text,        -- owner | admin | attorney | unknown
  bar_number  text,        -- the join key generic B2B data does not have
  linkedin_url text,
  phone_direct text,
  source      text, confidence numeric, enriched_at timestamptz
);

-- one contact may carry several addresses at different verification states
CREATE TABLE contact_email (
  contact_id  uuid NOT NULL REFERENCES contact,
  email       text NOT NULL,
  status      text NOT NULL,   -- valid|catch_all|risky|invalid|unverified
  pattern     text,               -- 'first.last' | 'flast' — feeds R3
  verified_at timestamptz,
  PRIMARY KEY (contact_id, email)
);


-- ═══ SIGNAL ═══════════════════════════════════════════

-- append-only. weight is NOT stored — it is derived at score time
-- from kind + age, so an ageing job posting gains weight without
-- the watcher having to emit a second event. see §9.
CREATE TABLE signal (
  id          bigserial PRIMARY KEY,
  firm_id     uuid NOT NULL REFERENCES firm,
  kind        text NOT NULL,
  fingerprint text NOT NULL,   -- dedupe key: same posting ≠ new signal
  first_seen  timestamptz NOT NULL,
  last_seen   timestamptz NOT NULL,
  closed_at   timestamptz,         -- posting removed → signal ends
  evidence    jsonb NOT NULL,   -- the quote the first line is built from
  UNIQUE (firm_id, kind, fingerprint)
);


-- ═══ LEAD ═══════════════════════════════════════════

-- a candidacy, not an entity. a firm may produce many over time.
CREATE TABLE lead (
  id            uuid PRIMARY KEY,
  firm_id       uuid NOT NULL REFERENCES firm,
  contact_id    uuid REFERENCES contact,
  grade         char(1),        -- 'A' verified email | 'B' phone-first
```

```
  state         text NOT NULL,
  wedge_sku     text,           -- from tier → sku lookup
  intent_score  int,
  arming_signal bigint REFERENCES signal,   -- what woke it
  cooldown_until timestamptz,
  created_at    timestamptz NOT NULL,
  closed_at     timestamptz, close_reason text
);
-- INVARIANT: at most one open lead per firm
CREATE UNIQUE INDEX ON lead (firm_id) WHERE closed_at IS NULL;

CREATE TABLE lead_state_event (
  lead_id     uuid NOT NULL REFERENCES lead,
  from_state  text, to_state text NOT NULL,
  trigger     text NOT NULL, actor text,
  payload     jsonb, at timestamptz NOT NULL
);

CREATE TABLE outreach_touch (
  id        bigserial PRIMARY KEY,
  lead_id   uuid NOT NULL REFERENCES lead,
  channel   text, step int, mailbox text,
  sent_at   timestamptz, delivered bool, bounced bool,
  replied_at timestamptz, reply_sentiment text
);


-- ═══ CONTROL ═══════════════════════════════════════

-- checked before EVERY send. domain scope is the important one:
-- one partner opting out silences the whole firm.
CREATE TABLE suppression (
  scope   text NOT NULL,    -- domain | email | firm_id
  value   text NOT NULL,
  reason  text NOT NULL,    -- customer | optout | dq | bounce
  added_at timestamptz NOT NULL,
  PRIMARY KEY (scope, value)
);
-- NOTE: 'competitor' is deliberately NOT a suppression reason.
-- a firm already buying from a rival is category-educated with an
-- approved budget — the strongest population there is. it goes on a
-- slower displacement track, not into silence.

CREATE TABLE firm_track (
  firm_id  uuid PRIMARY KEY REFERENCES firm,
  track    text NOT NULL DEFAULT 'cold',
                    -- cold | displacement | partner_intro | holdout
  reason   text, set_at timestamptz
);
-- 'holdout' carries the deliberate tier-4 and non-concentration
-- slices that exist to test targeting assumptions, not to convert.

-- §10 arming gate reads this. no bench → no outreach for that SKU.
CREATE TABLE bench_inventory (
  sku          text, language text,
  available    int, replenish_rate_30d int, as_of timestamptz,
  PRIMARY KEY (sku, language)
);
```

WHY `BENCH_INVENTORY` IS IN THE DATA LAYER

The product promises a start in under three days. If a lead is armed for a wedge SKU with an empty bench, the pipeline is manufacturing a promise the delivery side cannot keep — and the failure lands on the strongest part of the pitch. Making inventory a table the arming gate joins against turns that from an operational accident into a scheduling constraint.

§4 Normalization contracts

Identity resolution is only as good as the canonical forms it compares. Each function below is pure, deterministic, versioned, and must be applied at write time — never at query time, or the indexes are useless.

CANONICAL FORMS · ALL APPLIED BEFORE ANY COMPARISON

FUNCTION	RULE	FAILURE MODE IF SKIPPED
norm_domain	Lowercase → strip scheme, `www.` / `m.` / `amp.` → reduce to eTLD+1 via the Public Suffix List → drop if in `SHARED_HOST_BLOCKLIST` (wixsite, squarespace, godaddysites, justia, avvo, findlaw).	Every firm on a shared host merges into one entity. Silent and catastrophic.
norm_phone	Parse with libphonenumber, region `US` → E.164. Then: if one number resolves to **≥4** distinct firm candidates, mark it `shared` and exclude from matching — it is an answering service or a tracking pool.	Stored formatted, rule 2 never fires, dedupe silently degrades to name matching only.
norm_addr	USPS canonical abbreviations, secondary unit split into its own field (never in the hash — suites differ within one firm), ZIP+4 where available. `addr_hash = sha1(street_norm || zip5)`.	Suite numbers in the hash means one firm's two floors become two firms.
norm_name	Lowercase → strip punctuation → remove tokens `{law, laws, legal, office, offices, attorney, attorneys, firm, group, associates, pllc, pc, llp, llc, ltd, and, the}` → collapse whitespace. **Token order preserved** — trigram similarity depends on it.	Everything becomes "smith" and rule 4 matches every firm in the ZIP.
norm_practice	Map free text to a closed vocabulary of 15 slugs. Multi-area firms get an ordered array; `primary_practice` is the area with the most page real estate, not the first listed.	Tier assignment becomes non-deterministic across re-runs.

> **VERSION EVERY NORMALIZER**
>
> Store `norm_version` alongside each canonical value. When `norm_name` changes — and it will, the first time a firm called "The Law Offices of Law & Law" breaks it — you need to know which rows were computed under the old rule so they can be recomputed without re-crawling. This is a one-column decision that saves a full re-harvest.

§5 Identity resolution

One firm appears in Places, two bar directories and Justia under four different names. Two genuinely different firms in one city are called "Smith Law" and "Smith Legal". Getting this wrong in the safe direction produces duplicates; getting it wrong in the unsafe direction silently deletes prospects.

```
-- Blocking, then rules, then union-find. Never O(n²) over 300k records.

STEP 1 — BLOCK
  candidate pairs are generated only within blocks:
    block_by_domain  : exact norm_domain
    block_by_phone   : exact norm_phone (excluding shared)
    block_by_zip     : zip5
  → reduces ~320k records to ~4M candidate pairs, not 51 billion

STEP 2 — RULES (evaluated in order, first match wins)
  R1  norm_domain equal, both non-null, neither blocklisted   → MERGE
  R2  norm_phone equal, both non-null, neither shared         → MERGE
  R3  addr_hash equal AND similarity(name_stem) > 0.55        → MERGE
  R4  same zip5 AND similarity(name_stem) > 0.82              → REVIEW

  R3 requires both conditions. Address alone merges every firm in
  an office tower. R4 never auto-merges — a false merge is invisible
  and unrecoverable; a false split is visible and costs one duplicate.

STEP 3 — UNION-FIND
  merge edges from R1–R3 → connected components → one firm per component
  canonical_name = the variant from the highest-precedence source (§8)
  every other variant is written to firm_alias with its merge_rule

STEP 4 — VERIFY (run before the pipeline advances, not after)

  SELECT domain, count(*) FROM firm
   WHERE domain IS NOT NULL GROUP BY 1 HAVING count(*) > 1;
  SELECT phone_e164, count(*) FROM firm
   WHERE phone_e164 IS NOT NULL GROUP BY 1 HAVING count(*) > 1;

  Both must return ~zero rows. Thousands means R1/R2 never fired —
  almost always because phone was stored formatted rather than E.164,
  or www./trailing slash was not stripped before hashing.

ACCEPTANCE
  collapse ratio (raw ÷ resolved) expected 2.0–3.3×.
  Above 3.3× → over-merging, check R3/R4 thresholds.
  Below 2.0× → under-merging, check normalizers, NOT thresholds.
```

> **A BROKEN DEDUPE PRESENTS AS SUCCESS**
>
> More rows, more coverage, better-looking counts. It surfaces ten weeks later when one managing partner has received four emails from four of your sending domains and reported all of them. The verification query above is the only cheap moment to catch it — run it as a gate, not as a report.

§6 Scraper register

Twenty-four components. Every one declares an idempotency key so a re-run is free, and a failure policy so a dead source degrades the pipeline rather than stopping it.

PLANE D — DISCOVERY · CREATES FIRM CANDIDATES

ID	COMPONENT	INPUT → OUTPUT	IDEMPOTENCY KEY	ON FAILURE
D1	**Places grid harvester** 15 practice keywords × ~1,200 towns	`grid_cell` → `observation` Text Search then Details for each new place_id	`(cell_id, keyword, page)`	Cell marked failed, retried next run. Never blocks other cells.
D2	**Bar directory harvester** Three archetypes, not fifty scrapers	`state` → `observation` + `contact` **A** bulk export · **B** paginated search · **C** per-record lookup only	`(state, bar_number)`	Per-state isolation. A blocked state does not stall the other 49.
D3	**Legal directory crawler** Justia, Avvo, Martindale — gap fill only	`practice+geo` → `observation` Runs only for geos where D1 coverage is thin	`(source, profile_url)`	Optional plane. Failure logs a coverage warning, nothing more.

PLANE P — PROFILE · FILLS FIRM ATTRIBUTES, ONCE PER FIRM

ID	COMPONENT	INPUT → OUTPUT	NOTES
P1	**Site mapper**	`firm.domain` → `url set`	robots.txt, then sitemap.xml, then path probe for `/about`, `/attorneys`, `/our-team`, `/practice-areas`, `/contact`, `/careers`, `/fees`.
P2	**Page fetcher**	`url set` → `document` (GCS)	Content-hash keyed. **Store raw HTML always** — a schema change later becomes a re-parse, not a re-crawl.
P3	**Render fallback**	`document` → `document`	Playwright, gated: fires only when static text is under 500 words. Expect 5–8% of sites.
P4	**Boilerplate stripper**	`html` → `main text`	trafilatura. Removes the footer that lists every practice area the firm ever mentioned — an accuracy fix, not just a size one.
P5	**JSON-LD extractor**	`html` → `firm_attribute`	Deterministic. `schema.org` `LegalService`, `Attorney`, `JobPosting`. Highest confidence of any source.
P6	**Fee-model detector**	`text` → `fee_model`	Regex, precedence-ordered. Patterns in the block below.
P7	**Tech fingerprinter**	`html` → `cms_detected`	Script hosts, chat widgets, form endpoints. Filevine, Clio, CASEpeer, MyCase, Litify, SmartAdvocate. No vendor API needed.
P8	**LLM extractor**	`text` → `firm_attribute`	`claude-haiku-4-5`, Batch API, structured outputs. Contract in §7.
P9	**LLM escalator**	`low-conf rows` → `firm_attribute`	`claude-opus-5`. Fires on low confidence, `UNCLASSIFIED` tier, or tier 1A regardless of confidence.

```
-- P6 fee-model patterns. precedence top to bottom; first match wins.

contingency   \bno\s+(?:fee|win|recovery)\b.{0,24}\b(?:unless|until|if)\b
              | \bcontingen(?:cy|t)[\s-]?fee\b
              | \byou\s+(?:pay|owe)\s+nothing\s+unless\b

fee_shifting  \bfees?\b.{0,20}\bpaid\s+by\s+(?:the\s+)?(?:defendant|employer|manufacturer)\b
              | \bfee[\s-]shifting\b
              | \b(?:they|the\s+other\s+side)\s+pay(?:s)?\s+(?:my|our|your)\s+fees\b

flat          \bflat[\s-](?:fee|rate)\b
              | \$[\d,]+\s+(?:flat|total|all[\s-]inclusive)\b

hourly        \bhourly\s+rate\b | \bbilled\s+(?:hourly|by\s+the\s+hour)\b
              | \bper\s+hour\b

-- A firm matching BOTH contingency and hourly is 'mixed', not 'contingency'.
-- Resolve by page real estate: whichever fee language appears on the
-- practice pages with the most content wins. If still tied → 'mixed'
-- → tier UNCLASSIFIED → P9 escalation. Never guess this field.
```

PLANE W — WATCHERS · EMIT SIGNALS, NEVER TERMINATE

ID	WATCHER	CADENCE	SCOPE	SIGNAL FINGERPRINT
W1	**Careers JobPosting watcher** The highest-value job in the system	Nightly	All qualified firms	`sha1(job_title + datePosted)` Own `first_seen` is what makes "open >21 days" computable at all
W2	**Job aggregator poller**	Nightly	Firms with no careers page	`sha1(source + posting_id)`
W3	**Review poller**	Weekly	Tier 1A + 1B	`sha1(review_id)` Phrase-matched for intake failure; stores the sentence
W4	**Ad library poller**	Monthly	Tier 1A	`sha1(page_id + month)`
W5	**Mass tort detector**	Monthly	Tier 1A	`sha1(firm + tort_slug)` Tort keyword set matched against ad copy and landing pages
W6	**Tech delta watcher**	Quarterly	All qualified	`sha1(cms_from + cms_to)` Only a *change* is a signal; steady state is not
W7	**Liveness prober**	Rolling 90d	All	— Content hash change → re-queue P2. Dead site → `STALE`

PLANE R — RESOLVERS · PEOPLE AND VERIFICATION

ID	COMPONENT	INPUT → OUTPUT	NOTES
R1	**Contact enricher**	`firm` → `contact`	Multi-vendor adapter behind one interface. Runs **only** on firms that passed the tier gate — enriching the whole universe is the largest avoidable cost in the system.
R2	**Bar record joiner**	`bar rows × firm` → `contact`	Matches D2 attorney records to resolved firms on name + address. Yields real partner names free, which generic B2B sources do not carry.
R3	**Email pattern inferencer**	`known-good @domain` → `candidate`	Learn the firm's pattern from any verified address, apply to unresolved contacts at the same domain. **Every inferred address still goes through R4.**
R4	**Email verifier**	`email` → `status`	SMTP verification. `catch_all` and `risky` are **failures**, not passes — treating them as valid is how a sending domain dies in week one.

ID	COMPONENT	INPUT → OUTPUT	NOTES
R5	**Phone normalizer + DNC scrub**	`phone → e164 + flag`	Business lines sit outside most DNC scope, but solos routinely list a mobile as the firm number. Scrub anyway.

TWO SOURCES ARE DELIBERATELY ABSENT

Direct job-board scraping and automated LinkedIn extraction are both excluded. W1 reads the firm's own careers page — the same signal, from the firm's own server, with a real `datePosted` — and W2 covers the remainder through a licensed aggregator. The signal is not lost; only the exposure is.

§7 Extraction contract

P8 and P9 share one schema. Three rules make the output trustworthy: **every field carries a confidence**, `unknown` **is a first-class value**, and **every value carries the span it was read from** — which turns QA from re-reading pages into sampling evidence strings.

```
// output_config.format — enforced schema, not a prompt instruction
{
  "primary_practice": { "value": enum15, "confidence": float, "evidence": string },
  "practice_areas":   [ { "area": enum15, "weight": float } ],

  "fee_model": {
    "value":      "contingency"|"fee_shifting"|"flat"|"hourly"|"mixed"|"unknown",
    "confidence": float,
    "evidence":   string          // the exact sentence. required if != unknown
  },

  "attorney_count": {
    "value":      int|null,
    "confidence": float,
    "method":     "team_page_count"|"stated"|"inferred",
    "excluded":   int             // staff bios NOT counted as attorneys
  },

  "staff_ratio":      { "value": float|null, "confidence": float },
  "decision_makers":  [ { "name": string, "title": string,
                          "role": "owner"|"admin"|"attorney", "evidence": string } ],
  "languages":        [ iso639 ],
  "spanish_intake":   { "value": bool, "evidence": string },
  "entity_kind":      { "value": "private_firm"|"inhouse"|"gov"|"directory_stub" },
  "has_inhouse_recruiter": { "value": bool, "evidence": string }
}

-- ESCALATION TO P9 (claude-opus-5) — any of:
   fee_model.confidence        < 0.60
   attorney_count.confidence   < 0.60
   primary_practice.confidence < 0.70
   fee_model.value IN ('mixed','unknown')
   icp_tier(firm) = '1A'            -- always, regardless of confidence

-- PROMPT VERSIONING
   firm_attribute.source = 'llm:p8:v3'  -- model + component + prompt rev
   A prompt change bumps the rev; rows at the old rev are re-runnable
   from stored HTML without re-crawling anything.
```

WHY `UNKNOWN` MUST BE REACHABLE

A model forced to choose between `contingency` and `hourly` will choose — confidently, and sometimes wrongly. Fee model is the primary tier axis, so a confident wrong value silently routes a firm into the wrong segment with the wrong wedge SKU and no error anywhere. Making `unknown` a legitimate answer converts a silent misclassification into an escalation.

§8 Attribute reconciliation

Places says the phone is X, the website says Y, the bar directory says Z. All three are stored in `firm_attribute`; `reconcile()` materializes one winner into `firm` and can always explain why.

SOURCE PRECEDENCE BY FIELD · HIGHEST FIRST

ATTRIBUTE	PRECEDENCE	RATIONALE
canonical_name	site > bar > places > directory	The firm's own masthead is what they call themselves; Places carries stale DBA names.
phone_e164	site > places > bar	Site contact page is current. Bar records lag by years.
address	places > site > bar	Places is geocoded and structured; site addresses are free text in a footer.
primary_practice	llm(site) > directory category > places category	Places categories are coarse and self-selected at listing time.
attorney_count	llm(team page) only	No other source is even approximately right. Absent team page → null, not a guess.
fee_model	regex(site) > llm(site)	Deterministic match on explicit fee language beats inference. Disagreement → escalate.
contact identity	bar > site > enrichment vendor	Bar records are authoritative on who actually practises at the firm.

```
FUNCTION reconcile(firm_id, attr):
  claims ← firm_attribute WHERE firm_id=? AND attr=?
           AND observed_at > now() - freshness_window(attr)

  IF claims is empty            → NULL, mark attr stale
  IF one source only            → that value
  IF sources agree              → that value, confidence = max
  IF sources disagree:
      winner ← highest precedence source for this attr
      IF winner.confidence < 0.6 AND attr is tier-determining
          → emit reconcile_conflict, set NULL, queue P9
      ELSE → winner.value

  -- conflicts are never silently resolved for fee_model,
  -- primary_practice or attorney_count. those three decide the tier.

TRIGGER: any INSERT on firm_attribute → reconcile(that attr)
           → if value changed → recompute fit_score → re-evaluate tier
           → if tier changed → emit lead state transition (§10)
```

§9 Scoring engine

Two scores, computed differently and never merged before routing. **Fit** is a pure function of firm attributes — deterministic, recomputed on attribute change. **Intent** is a windowed aggregation over `signal` — recomputed nightly, decaying.

```
-- FIT — pure, 0..50, recomputed on attribute change
  fee_model IN (contingency, fee_shifting, flat)  +15
  practice 1A / 1B / 2 / 3                        +10 / +8 / +8 / +5
  attorney_count 2–12 / 13–25 / 1                 +12 / +8 / +5
  country = US                                     +6
  review_count > 40        -- case-volume proxy       +4
  staff_ratio < 1.0        -- understaffed            +4
  paid_ads_active          -- cash stability proxy     +3
  spanish_intake                                   +5
  attorney_count 26–50                            −10


-- INTENT — 0..50, nightly, weight derived at score time

-- weight is NOT stored on the signal row. an open job posting
-- gains weight as it ages — the same event, re-weighted, not a
-- second event. this is why signal has first_seen/last_seen/closed_at
-- rather than a static weight column.

FUNCTION signal_weight(kind, first_seen, closed_at) →
  CASE kind
    WHEN 'job_posting_open' THEN
        CASE WHEN closed_at IS NOT NULL        THEN  0
             WHEN age(first_seen) > 21 days    THEN 40   -- hiring has failed
             WHEN age(first_seen) >  7 days    THEN  34
             ELSE                                    30 END
    WHEN 'partner_intro'          THEN 35
    WHEN 'pricing_page_visit'     THEN 25
    WHEN 'review_unreturned_call' THEN 15
    WHEN 'multi_opening'          THEN 10
    WHEN 'paid_ads_spike'         THEN 10
    WHEN 'mass_tort_campaign'     THEN 10
    WHEN 'new_office_or_lateral'  THEN  8
    WHEN 'cms_implemented'        THEN  8
  END

FUNCTION decay(days_since_last_seen) →
  ≤7d → 1.00   ≤30d → 0.70   ≤90d → 0.35   >90d → 0

intent = LEAST(50, SUM(signal_weight × decay))  GROUP BY firm_id


-- CONTEST LEVEL — how many competitors see this signal
-- routing input, not a score component. a job posting is the
-- strongest intent signal AND the one every rival sees the same
-- day; a review-mined signal is weaker but uncontested. spend
-- human SDR time on high-intent, LOW-contest leads.
  job_posting_open        → contest HIGH
  paid_ads_spike          → contest HIGH
  review_unreturned_call  → contest LOW
  mass_tort_campaign      → contest MED
  cms_implemented         → contest LOW
  partner_intro           → contest NONE


-- ROUTING BAND
  fit + intent ≥ 80 AND contest ≤ MED   → STRIKE     (human, same day)
  fit + intent ≥ 80                     → SEQUENCE   (automated, 9 touches)
  fit + intent ≥ 60                     → SEQUENCE
```

```
    fit + intent ≥ 40                    → NURTURE
    otherwise                            → HOLD
```

§10 Lead lifecycle

A firm is built once. A lead is armed, worked, and returned to the pool — repeatedly, for years. The loop is the system: most firms are not in market today, and the pipeline's job is to be holding a verified, current record on the day they become in market.

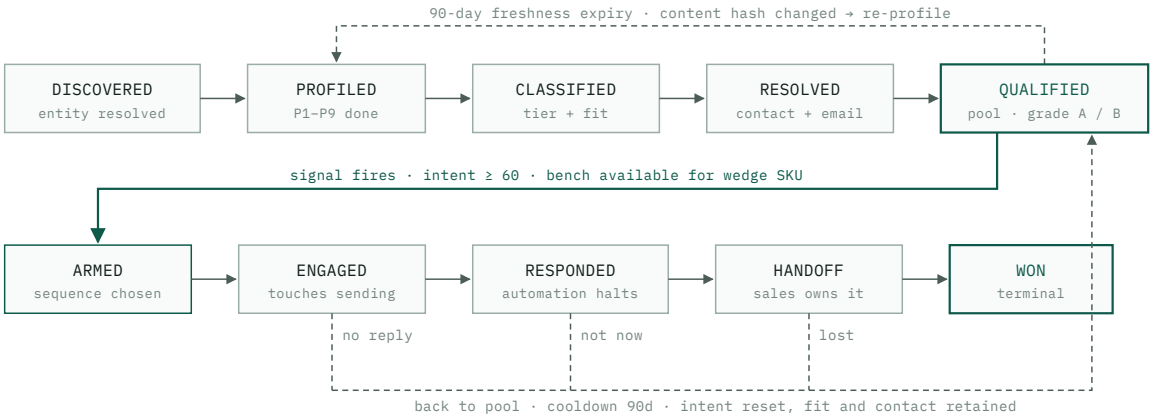


Fig 2 — The spine runs once per firm; the cycle runs indefinitely. The dashed return is the mechanism that matters: a lead that does not convert is not lost, it is re-pooled with its verified contact intact, waiting for the next signal. Terminal exits (`DISQUALIFIED`, `SUPPRESSED`) are omitted here for legibility and are exhaustive in the table below.

Transition table

Sixteen states. Every transition names its trigger, its guard, and what it writes — a transition with no guard is a bug waiting to send mail to a suppressed domain.

COMPLETE TRANSITION SET · GUARDS ARE ENFORCED IN THE DB, NOT THE SEQUENCER

FROM	TO	TRIGGER	GUARD	WRITES
—	**DISCOVERED**	Identity resolution assigns a firm id	Component has ≥1 observation	`firm`
DISCOVERED	**PROFILING**	P1 enqueued	`domain IS NOT NULL`	`crawl_job`
DISCOVERED	**NO_WEB**	No resolvable site	—	Held; D3 gap-fill may revive
PROFILING	**PROFILED**	P4–P8 complete	All required attrs present or explicitly null	`firm_attribute`
PROFILED	**CLASSIFIED**	`reconcile()` then `icp_tier()`	Tier ≠ `UNCLASSIFIED`	`firm.tier, fit_score`
PROFILED	**ESCALATED**	Tier is `UNCLASSIFIED` or confidence low	—	P9 queue
ESCALATED	**CLASSIFIED**	P9 returns	Tier resolved	`firm_attribute`
CLASSIFIED	**DISQUALIFIED**	Tier is `DQ_*`	—	`lead.close_reason` Revisitable: an attribute change re-runs the tier
CLASSIFIED	**CONTACT_PENDING**	Tier passes	Domain not suppressed	R1/R2 queue

FROM	TO	TRIGGER	GUARD	WRITES
CONTACT_PENDING	**RESOLVED**	R1/R2/R3 return a named person	`role IN (owner, admin)`	`contact`
RESOLVED	**QUALIFIED_A**	R4 verifies	`status = 'valid'`	`lead` grade A
RESOLVED	**QUALIFIED_B**	R4 fails, phone present	`status IN (catch_all, risky)`	`lead` grade B Phone and LinkedIn motion only. Never emailed.
QUALIFIED_*	**ARMED**	Nightly scorer crosses band	**intent ≥ 60** *and* not in cooldown *and* `bench_inventory` has the wedge SKU	`lead.arming_signal`
ARMED	**ENGAGED**	Sequence starts	Suppression re-checked at send time, not arm time	`outreach_touch`
ENGAGED	**RESPONDED**	Any inbound reply	—	Automation halts immediately
ENGAGED	**QUALIFIED_***	Sequence exhausted, no reply	—	`cooldown_until = +90d`; intent reset, contact retained
RESPONDED	**HANDOFF**	Positive reply	—	Owner assigned
RESPONDED	**QUALIFIED_***	"not now"	—	Cooldown from the date the prospect named
RESPONDED	**SUPPRESSED**	Opt-out	—	`suppression` at **domain** scope
HANDOFF	**WON / LOST**	Sales outcome	—	Attribution back to `arming_signal`
LOST	**QUALIFIED_***	Cooldown elapses	Loss reason not permanent	Re-pooled
QUALIFIED_*	**STALE**	W7 detects content-hash change or 90d expiry	—	Re-queue P2 → PROFILING
any	**SUPPRESSED**	Opt-out, hard bounce, or customer-list load	—	Sticky. Only a human clears it.

Invariants

Each of these is enforced by a constraint or a trigger, not by convention. Anything enforced only in application code will eventually be bypassed by a backfill script.

#	INVARIANT	ENFORCED BY
I1	At most one open lead per firm.	UNIQUE INDEX ON lead (firm_id) WHERE closed_at IS NULL
I2	No `outreach_touch` may be written for a lead whose firm domain is in `suppression`.	BEFORE INSERT trigger — re-checked at send, not at arm
I3	Grade B leads may never be emailed.	CHECK (grade='A' OR channel <> 'email')
I4	A lead cannot enter `ARMED` without bench inventory for its wedge SKU.	Arming query joins bench_inventory; no join, no row
I5	Every state change writes exactly one `lead_state_event`.	AFTER UPDATE trigger on lead.state
I6	`signal` is append-only; correction is a new row with a later `last_seen`.	REVOKE UPDATE, DELETE

#	INVARIANT	ENFORCED BY
I7	Every `firm` value traces to at least one `firm_attribute` row.	reconcile() is the only writer of those columns

§11 Orchestration

Prefect flows on Cloud Run Jobs. Two shapes: **one-shot flows** that walk a work queue until empty and terminate, and **cron flows** that never terminate.

```
ONE-SHOT — run once per firm, resumable, idempotent

  harvest_universe      D1 → D2 → D3        (parallel by state/cell)
       ↓
  resolve_identity      §5, full-set operation, gate: verify queries
       ↓
  profile_firms         P1 → P2 → P3? → P4  (fan-out, 1 req/2s per domain)
       ↓
  extract_attributes    P5 → P6 → P7 → P8 → P9?   (P8 batched)
       ↓
  reconcile_and_class   §8 → icp_tier()     gate: tier distribution sane
       ↓
  resolve_contacts      R2 → R1 → R3 → R4   (tier-gated: qualified only)
       ↓
  emit_leads            §10 QUALIFIED_A / QUALIFIED_B


CRON — never terminate

  02:00  W1  careers watcher        all qualified firms
  02:30  W2  aggregator backfill    firms with no careers page
  03:00  score_nightly              fit + intent + band, full scan
  03:30  arm_leads                  band crossings → ARMED (bench-gated)
  04:00  sequence_dispatch          ARMED → ENGAGED, respects send caps
  Sun    W3  review poller          tier 1A + 1B
  1st    W4/W5 ads + mass tort      tier 1A
  Qtr    W6  tech delta             all qualified
  rolling W7 liveness               1/90th of universe per day


IDEMPOTENCY
  every task keys on (component, entity_id, input_hash)
  a re-run with unchanged input is a no-op that logs and exits
  → any flow can be replayed from any point without duplication

BACKFILL
  a prompt or normalizer version bump does NOT re-crawl.
  it re-reads stored HTML from GCS and re-runs the affected
  component only. this is the single reason P2 stores raw HTML.

RATE DISCIPLINE
  per-DOMAIN token bucket in Redis, not global.
  one firm's site sees ≤1 request / 2s regardless of worker count.
  paid APIs get a separate bucket sized to the plan's QPS.
```

> THE GATE THAT MUST NOT BE ADVISORY
>
> `resolve_identity` is followed by the §5 verification queries as a **hard gate**. If firms share a domain or a phone, the flow stops. Everything downstream — crawling,

> extraction, enrichment, verification — is spent per-firm, so advancing past a broken dedupe multiplies the cost of the error by every subsequent stage.

§12 Observability

Most of these alert on **absence**, which is the failure mode that silently kills scraping systems. A watcher that returns zero results does not error; it just quietly stops feeding the pipeline.

ALERTS · EVERY ONE IS A DEFECT, NOT A METRIC TO WATCH

SIGNAL	CONDITION	WHY IT IS A DEFECT
Watcher silence	W1 finds 0 new postings across all qualified firms	Across thousands of firms, zero is statistically impossible. The parser broke or the fetch is being blocked.
Dedupe drift	count of firms sharing domain or phone > 0	Rules R1/R2 stopped firing. Duplicate outreach follows within days.
Extraction confidence drop	median `fee_model.confidence` falls >0.1 week over week	Site layouts changed, or a prompt revision regressed. Tier assignment degrades silently.
Tier distribution shift	any tier's share moves >15% week over week	Either the classifier drifted or the harvest changed shape. Both are bugs.
Bounce rate	>2% on any sending domain	Verification is leaking. Pause the domain; do not push through.
Signal-to-contact latency	p50 > 48h for STRIKE band	The one metric the whole signal architecture exists to minimise.
Bench starvation	a wedge SKU at 0 availability for >7 days	Leads for that tier are silently not arming. Looks like a demand problem; is a supply problem.
Stale share	>20% of pool past 90d freshness	W7 is behind. The pool is degrading toward an unusable list.

WHAT EACH STAGE MUST EMIT

- **Every fetch** → url, status, bytes, content hash, duration, proxy used, robots decision.
- **Every extraction** → component, model, prompt rev, input tokens, per-field confidence.
- **Every state transition** → the `lead_state_event` row. This table is the audit log; nothing else needs to be.
- **Every send** → mailbox, step, suppression check result, delivery outcome.

> **ATTRIBUTION CLOSES THE LOOP**
>
> Every `WON` and `LOST` carries its `arming_signal`. After ninety days that join — outcome by signal kind — is what re-weights §9. The weights in this document are a seeded hypothesis; the only thing that makes them correct is running that query and rewriting the table.

Build specification for the data layer only. Commercial modelling, funnel projections and cost estimates are deliberately out of scope.

Scoring weights, yield expectations and precedence orders are seeded values to be replaced by measurement. Source selection reflects terms-of-service posture as well as data quality; the two exclusions in §6 are deliberate.